Whole-component witnesses and a dimension bound of57 for finite-volume hyperbolic Coxeter polytopes

Internal manuscript translation of the exact proof atlas

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Abstract

We establish dimension at most57 for finite-sided, finite-volume real hyperbolic Coxeter polytopes. The final step is a local payment transfer valid specifically in ambient dimension58. Three whole rank15 components force a rank-capacity obstruction in the finite selected cases and a rigid energy-one spectator configuration in a genuine older ideal route. Two narrowly witnessed compensations then lower the exact finite charge maximum to377, equal to its curvature. Every ideal vertex retains surplus at least36, while the global two-extrema correction is4205. Their strict inequality excludes dimension58, and the completed internal bound58 supplies the higher-dimensional tail. We expose the complete finite and ideal certificate interfaces, the arbitrary-support proof and the precise boundary between published inputs, internal mathematics and partial proof-assistant verification.

1 Statement and proof architecture

Theorem 1. *Every finite-sided, finite-volume real hyperbolic Coxeter polytope has dimension at most57.*

No arithmeticity, compactness, simple-cusp or facet-count hypothesis is imposed. This is an internally reviewed computer-assisted mathematical proof. External refereeing, novelty, optimality and end-to-end Lean verification are not asserted. The exact endpoint is `fresh_d13_context_dimension57_20261018`; its recursive source manifest identifies the immutable proof atlas.

We prove the final step from an internally completed bound58, whose separate manuscript and certificate establish the higher tail. In particular, we do not infer that tail merely from excluding dimension58, nor assume an unproved downward passage to Coxeter facets. The preceding local-payment atlas and its complete arithmetic certificate are internal parts of the argument, not additional external assumptions.

For a two-face$F$ with$s(F)$ sides, write

$$\mathcal{D}_5(P) = \sum_F (5 - s(F)).$$

A simultaneous nonnegative corner rule paying at least2 on each triangle and1 on each quadrilateral gives an upper bound for$\mathcal{D}_5$. The final rule has finite-vertex charge at most377. At an ideal vertex its actual chargeC_{act} satisfies$\kappa - C_{\mathrm{act}} \geq 36$, where$\kappa$ is the link shape curvature. The global lower bound is

$$\mathcal{D}_5 \geq 377 v_f + \sum_{v \text{ ideal}} \kappa(v) + 4205.$$

The contradiction is therefore

$$0 \geq 4205 + 36v_i > 0.$$

The remainder proves the local and arithmetic interfaces that make this subtraction valid.

2 Actual corners, whole components and the previous finite count

We use outward norm-two roots, with diagonal Gram entries2 and nonpositive off-diagonal entries. At an actual two-face in dimension$n$, the supporting white roots form a positive-definite rank$n-2$ diagram. White roots together with one black side give a strictly positive-definite edge diagram; adjacent corners are positive semidefinite. A finite corner has an elliptic forest of rank$n$.

A whole component is a connected component of the full actual corner diagram, not merely of a displayed core. Presence conditions in a weight refer only to such whole components. This distinction is essential when additional supporting whites are restored.

At an ideal vertex with affine factor ranksr_i, the link is $\prod_i \Delta^{r_i}$, with$\sum r_i = n-1$. A marked pair in factor$i$ occurs in$\prod_{j\neq i}(r_j+1)$ actual corner fibers. Hence an affine factor envelope$(r+1)c(r)$ gives

$$C_{\mathrm{act}} \leq C_{\mathrm{up}} = V \sum_i c(r_i), \qquad V = \prod_i (r_i+1). \tag{1}$$

The factor-fiber theorem includes nonsimple ideal vertices. Whole affine factors used locally are established by the positive kernel and actual corner semidefiniteness, before applying this multiplicity.

Proposition 2 (Frozen complete finite interface)**.** *The preceding simultaneous rule in ambient58 pays every triangle at least2 and every quadrilateral at least1. Every actual rank58 finite forest has charge at most377 except precisely the following eight formal forests: oneB_{13} orD_{13} together with any multiset of three A_{15} orD_{15} components. Each of those eight has charge378. All ideal weights admit the envelope of Section 5.*

Certificate and completeness argument. The frozen coupled count starts from the complete3298 fixed-vector/rank5 cover, retaining fixedA_{17} separately from residualD_{17} and separating theA/B_{12} andD_{12} interaction types. Simultaneous actual base and cross-cost domination covers exceptional finite types and the arbitrary-label rank-two family. Its two potentially increasing context families are bounded separately:759 rows for theA_1/D_5 upgrade, and 539 rows for theA_2/A_4 guarded family, with all14 costly rows refined through77 actual assignments. Both bounds are strictly below377.

The complementary complete39 costly classes have748 fixed actual-type fibers. All actualA_1, A_2 multiplicities are fixed explicitly. The remainingA_3 presence guard is split into noA_3 and one reservedA_3; the latter may contain moreA_3 without multiplying the presence bonus. All other context guards are fixed, and residual pair interactions vanish. The1549 fresh residual tables certify every allowed last-item inequality, unreachable rank and attaining predecessor. Induction on an item's removal therefore proves their upper bounds for arbitrary residual multisets. Reconstructing every winning forest verifies its full contextual cost. All38 other classes are at most377; the residual-zero rank13-plus-three15 class gives exactly the eight listed forests. Their charge is$78+3(100)$.

The complete source, saved tables and separate replay are in `fresh_d17_a16_coupled_ambient58_frontier_20261018`. This proof interface is not inferred from sampled maximizing forests. The written universal local atlas is also required; the Bellman certificate alone does not prove that its costs correspond to valid corner payments. □

A separate frozenB_{13} endpoint theorem removes pair$(0,12)$, with no new coupon. Its complete selected domain has no affected image, but that absence alone is insufficient: two older affine-B_{13}

singleton routes really consumed the endpoint unit. Their whole cores instead give distinct already-unitA_1/D_{12} short-leaf contributions. The other endpoint-like pair$(1, 12)$ needed by the currentD_{12} theorem is retained. Thus the fourB_{13} forests in Proposition 2 drop to377.

3 The dimension-specific final local rule

Theorem 3 (Three-rank15 transfer)**.** *In ambient dimension58, make these changes to the simultaneous rule:*

1. *In a whole finiteD_{13}, give pairs$(0, 11), (0, 12)$ weight$1/2$ instead of1 only if its actual corner has three distinct whole unmarked A_{15} orD_{15} components.*
2. *Add$1/2$ to wholeD_{28} pair$(0, 12)$ only if its corner has two such whole unmarked rank15 components.*
3. *Add$1/2$ between an endpoint of wholeA_{12} and the long endpoint of wholeD_{16} under the same two-component presence guard.*

All ideal weights and other finite weights remain. Every triangle still pays at least2 and every quadrilateral at least1.

Witnesses are not counted with multiplicity. The negative guard uses $13+3\cdot15 = 58$ ranks, so the corner has exactly those four components. The two positive guards similarly force exactly$D_{28}+2\cdot15$ or $A_{12} + D_{16} + 2 \cdot 15$. The theorem is not asserted in larger ambient dimensions.

3.1 A general inverse bound for attached forests

Lemma 4. *For a connected positive-definite Coxeter Gram matrix$B$ of rank$k$, with diagonal2 and every nonzero native magnitude at least1, $(B^{-1})_{ii} \geq k/(k + 1)$ for every$i$. If a finite forest of total rank$M$ has a nonnegative contact vector$u$ with at least one unit contact in every component, then$u^t B^{-1}u \geq M/(M + 1)$.*

Proof. Inverse matrices of these positive-definite symmetric$Z$-matrices are entrywise nonnegative. Delete the chosen root; let the remaining connected components have ranksk_j. Every component has a contact with the root. By induction and inverse positivity its contribution to the root's Schur projection is at least$k_j/(k_j + 1)$. Since

$$\sum_j \frac{k_j}{k_j + 1} \geq \frac{\sum_j k_j}{1 + \sum_j k_j} = \frac{k - 1}{k},$$

Schur complementation gives $(B^{-1})_{ii} \geq (2 - (k - 1)/k)^{-1} = k/(k + 1)$. The rank-one base is$1/2$. Larger native magnitudes or multiple contacts only increase the lower bound. Apply it in every contacted component of a forest and use the same sum inequality to prove the second assertion. □

3.2 Complete selected cases and arbitrary additional whites

The fresh complete target domain contains801 patterns,862D_{13} maps, 33 affected contexts and76 affected pair maps. The physical cores have norm-two roots; full Lorentz, white and all four edge signatures and actual corner classifications are reconstructed exactly. The native-domain completeness is inherited from the universal reduction, rather than claimed from these records alone.

There are five proper completions. Their missing fork leaf is genuinely new: old corner roots outside the displayedA_{12} are orthogonal to it. Its forced unit contact has inverse projection square6,12,30 or102, exceeding2. Additional same-end contacts cannot lower the bound. Mixed-end contacts give an independently paid short path. Thus these completions are impossible.

Four radical cores, numbered1985,2378,2575,2581, seal the entire displayed core. Indeed the whole selectedD_{13} forbids contacts on its nodes; every remaining contact position has a nonzero radical coefficient. A new white must pair to zero with the radical. Nonempty contacts therefore meet both signs and give a retained adjacent-black shortcut of length at most8. Otherwise all contacts vanish and the unmarkedA_4 is whole. It cannot coexist with the three rank15 witnesses.

Twelve contexts similarly have a small displayed block with no permitted new contact of projection at most2. It is whole: any indirect extension would have a first new root contacting it. The rank-capacity guard again excludes it.

In the final twelve contexts the old small factor$K$ has rank$s = 1, 2, 3$. Its unique permitted contact position$p$ has core-inverse square $r = 4/3, 3/2, 2$, respectively. The native gap excludes magnitudes greater than1; positive-end inverse bounds exclude two same-end contacts. Four cases also permit an opposite contact position$q$, with $J_{pp} = J_{qq} = 4/3$ and$J_{pq} = -4/3$. Two new roots usingp, q have residual squares2/3, so their full product is at most$-4/3 + 2/3 = -2/3$. Their forced native bond supplies the retained short path. Thus, without that escape, all new whites in a growing$K$ component contact the old core only at$p$; some may be orthogonal to it.

Let$Y$ contain all new whites in this whole component, including every indirect tail, and let$M$ be its rank. Every component of$Y$ contacts$K$. With contact vector$u$, the full Lorentz condition gives the positive semidefinite residual Schur matrix

$$G_Y - ruu^t \succeq 0, \qquad u^t G_Y^{-1} u \le 1/r.$$

Lemma 4 implies$rM/(M+1) \le 1$, hence $M \le 3, 2, 1$ respectively. In all cases the whole$K$ component has rank at most4, contradicting the guard. This controls arbitrary supporting whites, not just immediate neighboring roots.

3.3 The universal affine-D12 consumer

The old opposite-fork-leaf affine-D_{12} theorem genuinely used the newly lowered whole finiteD_{13} unit. It must be repaired separately. Number its affine factor$F$ of order13 with spine0 through8, short leaves 9,10 at0 and11,12 at8. Choose ideal marks$a = 9, b = 11$ and opposite black rootsc, d. Let$Z$ be the elliptic spectator forest, orthogonal to$F$, andx, y the nonnegative contacts ofc, d with$Z$.

An adjacent-black bond or a shared spectator neighbor is an independently paid short path. Otherwisex, y have disjoint supports. The complete all-label end-profile reduction gives each end cost at most8; a nonextreme end has cost at most4. The exact Schur identity is

$$\frac{\det S}{h^2 k^2} = 1 - \frac{g_c + g_d}{12} + \frac{1}{12}(x/h - y/k)^t G_Z^{-1}(x/h - y/k).$$

Its determinant vanishes at the actual four-black star.

If an end is nonextreme, equality forces costs8 and4 and$x = y = 0$. A quiet root attached to the selected old leaf gives the retained whole A_1/D_{12} short-leaf unit. If it attaches to the mate, it gives whole D_{13}. Without the new guard its unit is unchanged. With the guard, Z consists of exactly three orthogonal whole rank15 factors. The three nonextreme equality profiles give, at a distinct corner, respectively a retainedA_1/D_{12} unit, an independent whole affine B_{13} priority, or a second guardedD_{13} half. These exhaust24 physical cases. The currentB_{13} endpoint repair uses itsD_{12} cross unit, not a newly financedD_{13} route; there is no circularity.

When both ends are extreme,$h = k = 1/2$, and the determinant becomes

$$(x - y)^t G_Z^{-1}(x - y) = 1. \tag{2}$$

The old two-nonempty-support theorem gives retained short paths or small sealed alternatives. The changedD_{13} output belongs to the arbitrary one-sided branch, say$y = 0$. The quiet corner excludes$c$ and retains$Z$ orthogonally. If itsD_{13} guard holds,Z is exactly three whole A_{15}/D_{15} factors.

Lemma 5. *Under the strict edge conditions and disjoint-support hypothesis, energy1 on three wholeA_{15}/D_{15} factors forces a single unit contact at the long endpoint of oneD_{15}, by one black root. The other two factors have zero contact with both black roots.*

Proof. For A_{15} the inverse diagonal entries below2 occur only at nodes 0,1,13,14. They are 15/16, 7/4, 7/4, and 15/16, respectively. Two native contacts give energy at least2 and violate strict edge positivity. The complete profiles are simple contacts at these four nodes or doubled contacts at0,14. The next native squared magnitude$(3 + \sqrt{5})/2$ is already excluded. ForD_{15} only its long endpoint has inverse diagonal below2, equal to1, and only its simple contact fits.

Every allowed nonzero component energy is at least15/16; noA_{15} profile has energy1. The finite exact inverse table also checks every disjoint two-sided profile difference: all are at least15/16 and none is1. Equal-node contacts are forbidden by disjoint support. Two active components would have total at least$15/8 > 1$. Equation (2) therefore forces precisely the statedD_{15} contact. □

The two inactive components are orthogonal to the entire star and the active component. They remain whole, without indirect bridges. Ifc attaches to selected$a$, the opposite corner is wholeA_{12} at an endpoint and wholeD_{16} at its long endpoint. If it attaches to$a$'s mate, that corner is wholeD_{28} at$(0, 12)$. Both physical30-root models have signature$(28, 1, 1)$, positive white rank26, positive edge rank27 and semidefinite corners. In either case the new half, guarded by the two inactive rank15 factors, supplements the quietD_{13} half at a different actual corner. This proves the required total1.

3.4 Chosen-output and triangle retention

The previous selected and nonselected routes must retain their chosen positive contributions, including every later replacement route. The low66 contexts have order bound at most12. The74 protected and9 adaptive paths have length9, whereas the changedD_{13} role has distance11. The current finite6/7/8 and cycle/tree routes terminate at short paths, protected leaves, small whole factors or unchanged ideal contributions.

The older affine-D_{11} repairs use$D_{13}(1, 11)/(1, 12)$ units, retained here. The current center/leaf theorem uses$D_{13}(0, 10)$, at its fork center. The coupled order11 unmixed boundary givesD_{11+r} marks $(r, n - 2)/(r, n - 1)$ of distance9; for$r = 2$ these are disjointD_{13} roles. The other current finite and contextual outputs are similarly retained. The new producer reconstructs all216 literal configurations,564 later corners,24D_{14} variants,332 positiveD_{12} outputs and the newer sealed families. These finite checks support the named interfaces; universal completeness remains the written routing theorem. The genuine unstored affine-D_{12} exception was repaired above.

For triangles use the complete actual-face theorem, not three pairs in one diagram. Low cores give two distinct corner paths of length at most5. The1025 high-core triples give paths of length at most7 or one of four protectedE_8 alternatives, with finite marked length6 or7 or whole exceptional affine order at most9. Current finite6 deletions retain all three exceptional length6 alternatives through their nearby whole leaf; the length7 alternative and other persistent-feature outputs survive. The present distance11 reduction changes none of these chosen payments. Ideal contributions are unchanged. This completes Theorem 3.

4 The new complete finite maximum

Proposition 6. *Every actual finite rank58 forest under the final simultaneous rule has charge at most377. The exact formal maximum is377.*

Proof. TheB_{13} theorem lowers four forests in Proposition 2 from378 to377. The newD_{13} guard lowers the other four by1. Every other forest was already at most377, except that the two new positive coupon families must be checked.

Rank capacity makes those families exact. They have respectively

$$170 + 2(100) + \tfrac{1}{2} = 370.5, \qquad 66 + 108.5 + 2(100) + 1 = 375.5.$$

The second total counts bothA_{12} endpoint coupons. The previous A_{12}/D_{16} cross cost is zero. Each family has three possible multisets of twoA_{15}/D_{15} factors, so all six actual forests are covered. No additional context witness can fit. Outside them the change is pointwise nonincreasing. Thus the bound377 holds everywhere, and the eight changed forests attain it formally. A geometric realization of an attainer is neither required nor asserted. □

5 Positive ideal surplus

For affine ranksr_i summing to57 set

$$H(t) = \prod_i (1 + t + \cdots + t^{r_i}) = \sum_{j=0}^{57} h_j t^j, \qquad V = \prod_i (r_i + 1),$$
$$G = \sum_j h_j \min(j, 57 - j), \qquad \kappa = \frac{5G - 57V}{2}.$$

The frozen all-family envelope has$c(r) = r/2$ for$r \le 8$ and $c(9) = 71/16$; the remaining exact rational entries are explicitly supplied in the certificate. They are unchanged by every final finite transfer.

Lemma 7 (Positive low8 reduction)**.** *Every actual ideal vertex has*$\kappa - C_{\mathrm{act}} \ge 36$.

Proof. For$1 \le a \le b \le 7$, replacing(a, b) by$(a - 1, b + 1)$ preserves normalized envelope cost. The centered sum of independent uniforms on $\{0, \dots, a\}$ and$\{0, \dots, b\}$ increases in convex order, verified by the complete28 exact stop-loss identities. Adding the remaining independent factors preserves this order. Since

$$\frac{G}{V} = \frac{57 - \mathbb{E}|2S - 57|}{2},$$

the transfer does not increase normalized curvature or normalized margin. Its product decreases from$(a + 1)(b + 1)$ to$a(b + 2)$. Iteration terminates, since the sum of squared ranks increases, at arbitrary ranks at least8 and at most one positive rank below8. Rank zero is omitted. No transfer passes through the changed rank9 envelope.

There are exactly2552 terminal profiles. Independent domain reconstruction and polynomial multiplication verify all of them. Their normalized margin$\kappa/V - \sum_i c(r_i)$ has minimum$87/10240 > 0$; their absolute envelope surplus has minimum36. If the original and terminal quantities are(V, m) and(V', m'), then$V \ge V'$ and$m \ge m' > 0$. Thus $Vm \ge V'm' \ge 36$. This multiplication uses positivity. Finally (1) transfers the same lower bound to actual cusp charge. □

6 Shape curvature and the contradiction

Lemma 8. *For a proposed ambient58 polytope,*

$$\mathcal{D}_5 \geq 377v_f + \sum_{v \text{ ideal}} \kappa(v) + 4205.$$

Proof. The published incidence/index reduction writes $\mathcal{D}_5 = \sum_v (5E_v^0/2 - E_v)$, where$E_v$ is the number of edges of the vertex figure andE_v^0 the edges not cut by its induced level section. A generic orientation gives the index distributionh_j and $E_v^0 \geq G_v$. At the distinct global minimum and maximum the whole vertex figure is uncut, and symmetry gives

$$E_v - G_v = \sum_{j>57/2} h_j(2j - 57) \geq \sum_{j=29}^{57} (2j - 57) = 841.$$

Every coefficient of a simplex or product-of-simplices$h$-polynomial is at least1. Adding the two extrema and multiplying by$5/2$ gives4205, counted once. For a finite vertex$h_j = 1$, so

$$G_0 = \sum_{j=0}^{57} \min(j, 57 - j) = 812, \qquad \frac{5G_0 - 57 \cdot 58}{2} = 377.$$

The remaining ideal terms are preciselyκ. □

For faces with at least five sides,$5 - s(F) \leq 0$. The nonnegative local rule and Proposition 6 therefore give

$$\mathcal{D}_5 \leq 377v_f + \sum C_{\text{act}}.$$

Subtract Lemma 8 and use Lemma 7:

$$0 \geq 4205 + \sum (\kappa - C_{\text{act}}) \geq 4205 + 36v_i > 0.$$

This also excludes the all-finite case. Thus dimension58 is impossible. The completed internal bound58 excludes all dimensions at least59, proving Theorem 1.

7 Reproducibility and assumption boundary

The final endpoint command is

```
python3 scripts/fresh_d13_context_dimension57_20261018.py --verify
```

It checks the frozen common bindings, the complete eight-obstruction and six-increasing-forest closure, all2552 polynomial products and margins, all28 stop-loss transfers and the constants377,4205. It does not repeat finite Bellman tables, native-support censuses or historical geometry. Those certificates have their own separate replay commands. The new local producer recomputes its33 contexts, exact inverse profiles, physical stars and marked-role interfaces; the written arbitrary-support argument remains part of its proof, not a finite absence test.

Published inputs are finite/affine Coxeter classification, the minimal indefinite order ceilings, standard finite-volume cusp/simple-edge facts, and the incidence/index formulas. The new local transfers, whole-factor seals, rank58 guards, typed-context budgets, factor-fiber multiplicity, convex-order reduction, strict extrema correction and completed dimension tails are internal obligations. They may not be imported as external axioms merely because their proofs are lengthy.

The Lean project is separate and partial. Its checked generic and executable arithmetic results do not by themselves formalize all geometry or the complete dimension theorem. The companion lemma map specifies quantified internal interfaces rather than axioms for these new results. No successful hash or TeX build is represented as a proof check.

A Minimal proof-atlas spine

Role	Artifact stem
Final57 composition	`fresh_d13_context_dimension57_20261018`
Dimension-specific local rule	`fresh_d13_three_rank15_context_transfer_20261018`
B13 endpoint repair	`fresh_b13_endpoint_retention_20261018`
Complete preceding finite count	`fresh_d17_a16_coupled_ambient58_frontier_20261018`
Current coupled local rule	`fresh_d17_leaf_a16_context_coupled_transfer_20261018`
Positive low8 cusp certificate	`fresh_order10_low8_ambient58_frontier_20261018`
Genuine ideal consumer	`fresh_affine_d12_leaf_pair_transfer_20261017`
Triangle protection	`fresh_finite7_ideal5_triangle_payments_20261017`
Current finite6 interface	`fresh_depth6_finite6_zero_transfer_20261018`
Completed higher tail	`fresh_a15_d15_dimension58_20261018`
Factor-fiber geometry	`fresh_cusp_factor_payment_framework_20261017`
Shape-curvature derivation	`fresh_cusp_link_shape_curvature_20261017`

The full recursive manifests, not this compact index, identify all prior local inputs. The quantified map and reproduction guide are `docs/finite_volume57_lemma_map.md` and `docs/finite_volume57_reproduce.md`.

References

[1] A. G. Khovanskii, *Hyperplane sections of polyhedra, toroidal manifolds, and discrete groups in Lobachevskii space* (1986), Theorems2,6,9 as cited in the shape-curvature report. `https://www.math.utoronto.ca/askold/1986-Faa-1-english.pdf`.

[2] M. Chein, *Recherche des graphes des matrices de Coxeter hyperboliques d'ordre*≤ 10 (1969), the classification and order ceiling cited by the high-rank census. `https://www.numdam.org/item/M2AN_1969__3_3_3_0.pdf`.

[3] A. Felikson and P. Tumarkin, *On simple ideal hyperbolic Coxeter polytopes*, arXiv:math/0502413, Section1 containment input, as distinguished from the internal factor-fiber proof. `https://arxiv.org/pdf/math/0502413`.